

Department of Physics, Princeton University

**Graduate Preliminary Examination
Part I**

Thursday, May 13, 2021
1:00 pm - 4:20 pm

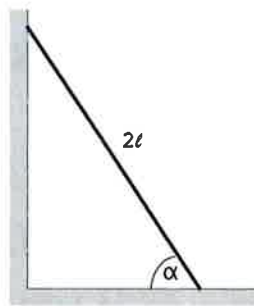
Answer TWO out of the THREE questions in Section A (Mechanics) and TWO out of the THREE questions in Section B (Electricity and Magnetism).

Work each problem in a separate examination booklet.

Be sure to label each booklet with your name, the section name, and the problem number.

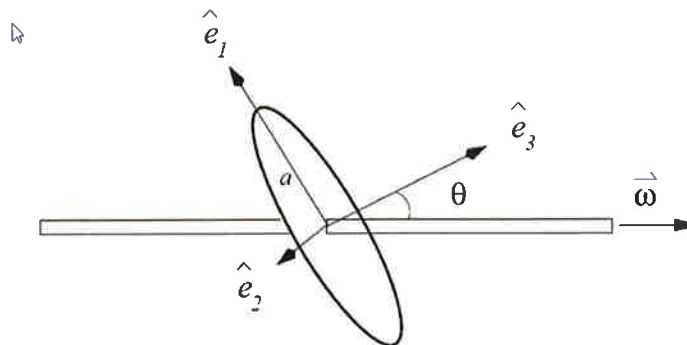
Section A. Mechanics

1. Slipping ladder



A uniform ladder of mass m and length $2l$ has one end on a smooth (frictionless) floor, and the other end against a smooth (frictionless) wall.

- (a) If the ladder is released from rest at an initial angle α with respect to the floor, find the angle at which the ladder leaves the wall.
- (b) Derive an analytic integral expression for the time elapsed between release and detachment, as a function of the initial angle (you can leave this expression in the form of an integral. You don't need to do the integration analytically).

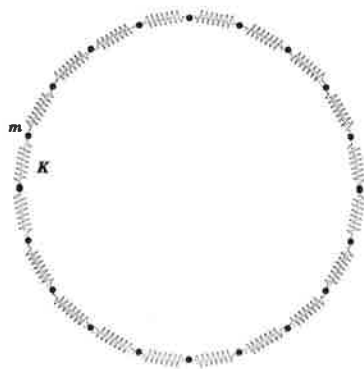


2. Rotating disk

A thin disk of mass m and radius a is attached to a massless axle through its center, see the figure. The axle is inclined with respect to the disk's symmetry axis ($\hat{e}_3$) by an angle θ . The disk is set to rotate with constant angular velocity $\vec{\omega}$ directed along the axle. Denote $\omega = |\vec{\omega}|$ in the rest of the problem.

Choose a set of body-fixed axes $\hat{e}_1$, $\hat{e}_2$, and $\hat{e}_3$ as shown in figure, where $\hat{e}_3$ is the symmetry axis of the disk, $\hat{e}_1$ is the principal axis of the disk lying in the plane defined by $\hat{e}_3$ and the axle, and $\hat{e}_2 = \hat{e}_3 \times \hat{e}_1$ (note that $\hat{e}_2$ is always orthogonal to the axle of rotation). The origin is at the center of mass of the disk.

- Find the inertia tensor of the disk in the body-fixed axes defined above.
- Find the kinetic energy of the disk.
- Find the components of the angular momentum in the body-fixed frame defined above.
- What are the components of the torque (in the body-fixed frame) that has to be applied in order to keep the disk from wobbling (i.e., in order to maintain the constant angular velocity along the axle)?



3. Periodic chain of masses and springs

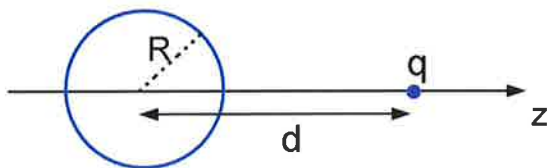
Consider N identical masses sliding without friction on a stationary circular hoop whose circumference is Na . The masses are connected by springs of spring constant K and natural length a , as shown in the next figure. Gravity is negligible.

The aim of this problem is to figure out all the normal modes and their frequencies.

- (a) Write down the Lagrangian and the equations of motion for this system. Use coordinates $q_1, q_2, \dots, q_N$ describing the displacements of the particles on the hoop from their equilibrium positions.
- (b) Show that the “plane wave” ansatz, $q_j = \text{Re}\{e^{ikaj+i\omega t}\}$, is a solution to the equations of motion, provided the frequency satisfies $\omega^2 = \frac{4K}{m} \sin^2(ka/2)$. Here k is supposed to be any real number, but we’ll restrict it in the next part of the problem.
- (c) Only certain values of k are permissible. What are they?
- (d) Some values of k are physically indistinguishable from others, in that they result in the same ω and the same $q_j(t)$. How many physically distinguishable values of k are there? Write down the expression for the corresponding eigenfrequencies. Is there any zero mode? If so, what is its physical interpretation?
- (e) Consider the continuous limit defined by $N \rightarrow \infty$, $a \rightarrow 0$, with the total length $\ell = Na$, total spring constant $\kappa = K/N$ and mass density $\sigma = m/a$ kept fixed. Write down the partial differential equation satisfied by the displacement $q(x, t)$ in this continuous limit, where x is the position along the hoop, $0 \leq x \leq \ell$. Write down what is the relevant boundary condition that $q(x, t)$ has to satisfy.

Section B. Electricity and Magnetism

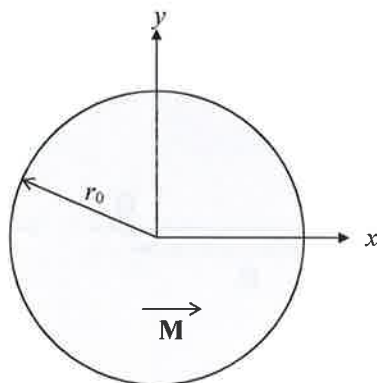
1. Point charge and conducting sphere



A point charge is at a distance d from the center of a conducting sphere of radius R . You may assume that $d > R$, as shown in the figure. The sphere carries no net charge (it is not grounded).

- Find the electrostatic potential everywhere. Use spherical coordinates r, θ, φ , where r is the distance from the center of the sphere, θ the angle with respect to the z axis in figure, and φ the angle around the z axis. [*Hint*: you may find it convenient to use the image charge method].
- What is the induced electric dipole moment of the sphere?
- Find the induced charge density on the surface of the sphere.
- What is the net force acting on the conducting sphere due to the point charge? Is the force attractive or repulsive?

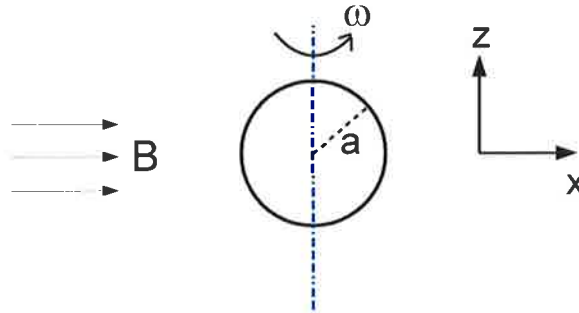
2. Moving Bar Magnet



A long cylinder of radius r_0 has uniform permanent magnetization density $\mathbf{M}$ perpendicular to the axis of the cylinder. Let $\hat{z}$ =axis of the cylinder, and $\mathbf{M} = M\hat{x}$ as shown in the figure.

- (a) Find the fields $\mathbf{B}$ and $\mathbf{H}$ everywhere.
- (b) Suppose the cylinder is given uniform velocity $\mathbf{v} = v\hat{z}$ along its axis. Find the resulting charge density and electric field everywhere. You may ignore effects of order $(v/c)^2$.

3. Rotating coil in a magnetic field



A circular conducting coil of radius a , resistance R and self-inductance L rotates with constant angular velocity ω along the z axis in the presence of a uniform magnetic field B directed along the x axis (see the figure).

- Find the steady state electric current $I(t)$ induced in the coil.
- What is the external torque $\vec{\tau}_{\text{ext}}$ (magnitude and direction) that has to be applied in order to maintain the constant rotation of the coil?
- Due to the time-dependent current, the coil radiates electromagnetic waves. What is the frequency of the emitted radiation?
- What is the total power radiated by the coil?

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**Graduate Preliminary Examination
Part II**

Friday, May 14, 2021
1:00 pm - 4:20 pm

Answer TWO out of the THREE questions in Section A (Quantum Mechanics) and TWO out of the THREE questions in Section B (Thermodynamics and Statistical Mechanics).

Work each problem in a separate booklet. Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Quantum Mechanics

1. Charged particle in a uniform magnetic field

Consider a spinless charged particle of mass m and charge q moving nonrelativistically in vacuum in three dimensions in a spatially uniform nonzero magnetic field $\vec{B}$. Choose the z axis to point along $\vec{B}$.

- (a) Choose a gauge for the electromagnetic vector potential, and write down the Hamiltonian for the particle.
- (b) Find the commutation relations between the components of the velocity operator of the particle. What does the result imply about measurements of the particle velocity?
- (c) Find a complete set of good quantum numbers, appropriate for your gauge choice, for the energy eigenstates of the particle, and all the corresponding eigenenergies. Be sure to include the motion along all three directions.
- (d) Write out the corresponding wavefunctions in terms of standard functions (you may leave them un-normalized).

2. Approximation methods for the absolute value well

A point mass m moves in one dimension under the influence of the potential

$$V(x) = F|x| \quad -\infty < x < \infty,$$

where F is a positive constant.

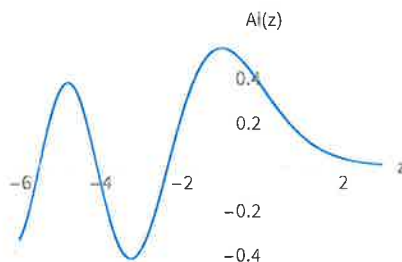
- Use the Schrodinger equation and dimensional analysis to find how the energy levels scale with the constant F .
- Use the WKB approximation to estimate the energy levels E_n , $n = 0, 1, 2, \dots$ of the particle in this potential well.
- Use the variational method with a Gaussian trial wavefunction to estimate the ground state energy E_0 . Compare the result to that of the WKB approximation. Which method do you expect to give a better approximation of the ground state energy? For this part, you may find the following integrals useful

$$\int_0^\infty dx e^{-x^2} = \frac{\sqrt{\pi}}{2}, \quad \int_0^\infty dx x e^{-x^2} = \frac{1}{2}, \quad \int_0^\infty dx x^2 e^{-x^2} = \frac{\sqrt{\pi}}{4}$$

- Now you may try to find the exact energy levels in this potential. Recall that the solution of the differential equation

$$\frac{d^2}{dz^2} f(z) = z f(z)$$

that decays at $z \rightarrow +\infty$ is given by the Airy function $\text{Ai}(z)$. This function has a series of zeros $\text{Ai}(z) = 0$ at $z = -2.3381, -4.0879, \dots$ and a series of critical points $\text{Ai}'(z) = 0$ at $z = -1.0188, -3.2482, \dots$ (see the figure below). Using these facts, find the exact values of the energies for the first four energy levels.



3. Bound states in double delta function potential

A particle with mass m moves in one dimension in the presence of the potential

$$V(x) = -\frac{\hbar^2 \beta}{m} (\delta(x+a) + \delta(x-a))$$

where β is a positive constant with dimension of inverse length. As you will show below, if the dimensionless quantity βa is large enough, this potential has two bound states: the ground state with energy E_0 and one excited state with energy E_1 .

- (a) Write down a suitable ansatz for the ground state wavefunction, and use the appropriate matching conditions at $x = \pm a$ to derive a transcendental equation that determines the ground state energy E_0 . Show that this equation always has a solution.
- (b) Repeat the above for the case of the excited state wavefunction, and derive the equation that determines the energy E_1 of the excited state. Show that this equation only has a solution above a certain critical value of βa . Find this critical value.
- (c) In the limit $\beta a \gg 1$, show that the two energy levels are exponentially close

$$E_1 - E_0 \approx C (e^{-2\beta a} + O(e^{-4\beta a}))$$

and find the constant C explicitly.

Section B. Statistical Mechanics and Thermodynamics

1. Molecular hydrogen

Molecular hydrogen H_2 consists of two hydrogen atoms linked by a covalent bond. Each hydrogen atom consists of one proton and one electron. The two electrons of H_2 pair up their spins so the net electronic spin is 0. Each proton has a magnetic moment, which is associated with the fermion's spin of $1/2$. In the H_2 molecule, the nuclear spins of the two protons couple to form a parallel triplet state known as orthohydrogen (spin $S=1$), and an anti-parallel singlet state known as parahydrogen (spin $S=0$).

- (a) Write down the partition function for the two spin states of H_2 . Assume the energy difference between the $S=0$ and $S=1$ states is Δ , and that the $S=0$ state lies lowest.
- (b) At room temperature, hydrogen is 70% $S=1$ and 30% $S=0$. Calculate a value for Δ from these data. (Remember: at room temperature $k_B T \approx 4 \times 10^{-21} \text{ J}$.)
- (c) The latent heat ΔL of hydrogen at its boiling point (20.3 K at 1 bar) is 445 kJ/kg. The conversion of $S=1$ to $S=0$ is very slow (several days), and it is quite possible to quickly cool hydrogen gas to its boiling point with the room temperature spin distribution intact. Assume $\Delta = 10^{-21} \text{ J}$. Determine if liquid H_2 will entirely boil off when the spin conversion to the ground state occurs. Assume perfect thermal insulation, and that the hydrogen goes completely to the ground state.

2. Spin-waves in a ferromagnet

Consider spin-waves in an isotropic three-dimensional ferromagnet where a spontaneous magnetization (which can point in an arbitrary direction) is present at temperatures below the “Curie temperature”.

The spin-waves (“magnons”) are collective modes in which the spins on each atom oscillate in space and time, and propagate as waves. As in the case of quantized sound waves (“phonons”), the spin waves can be treated at low temperatures as quantized harmonic-oscillator modes, and can store internal energy in a crystal lattice. However, these “magnons” have a different relation between frequency and wave number than the linear dispersion-relation of phonons. In particular, at low wavenumber

$$\omega(k) = Ak^2$$

where A is a constant. This is valid for $ka \ll 1$, where a is the lattice-spacing length scale.

Consider a magnetic material with ferromagnetically-ordered spins in thermal equilibrium at a low temperature T .

- (a) What is the average thermal energy in a spin-wave mode of frequency ω ? (Neglect the zero-point energy of the mode).
- (b) At low temperatures, the heat capacity of the spin wave system in the crystal is proportional to T^α . What is the numerical value of α ?
- (c) If the material is a metal (with a Fermi surface), do the spin waves give the dominant contribution to the heat capacity in the low-temperature limit? What if the material is an insulator? (Phonons are present in both cases.) Explain both of your answers.
- (d) If the ferromagnetic material was not a three-dimensional crystal, but is either (a) a metallic two-dimensional layer or (b) a one-dimensional metallic line of atoms (embedded in an insulating three-dimensional background), which contribution (metallic conduction electrons or magnons) dominates the heat capacity at low temperatures?

3. Specific heat and fluctuations

The energy E of a system in equilibrium with a thermal bath at temperature T is fluctuating continuously with $\text{Var}(E) = \langle E^2 \rangle - \langle E \rangle^2$ where $\langle E \rangle$ is the mean energy.

- (a) Show that the specific heat of the system at constant volume ($C_v = \frac{d\langle E \rangle}{dT}$) is directly related to the energy fluctuations by $\text{Var}(E) = T^2 C_v$.
- (b) Calculate the relative amplitude of the fluctuations

$$\delta = \frac{\sqrt{\text{Var}(E)}}{\langle E \rangle}$$

for an ideal gas of N molecules at temperature T .

- (c) Let's use this fundamental result about fluctuations to estimate the accuracy δ of a small detector to measure concentration. Suppose that a sensor of linear size a measures the concentration c of some molecules. The sensor counts the molecules within a volume a^3 at some instant. On average, the sensor counts $N_{ave} = ca^3$. Assume that the measurements of N are Poisson distributed. What is the measurement accuracy δ in that case?
- (d) An obvious strategy for the sensor to improve δ is by performing multiple measurements. The integration time τ has to be longer than the time it takes for molecules to diffuse in and out of the sensor volume. Estimate the time τ_0 it takes for a molecule with diffusion constant D to diffuse through the sensor volume, and use this to estimate how many independent measurements the sensor can perform in some integration time τ , and thus provide an expression for an improved accuracy δ .

